Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Interaction

Meaning in my words How would you explain interaction to a friend?	**How can you tell?** In “the rope pulls the crate,” which two objects are interacting?
Example or sketch What interaction holds up a book resting on a table?	**Non-example and why** Someone names “moving right” as the object pushing a cart. What would you ask?

Contact force

Meaning in my words How would you explain contact force to a friend?	**How can you tell?** Which detail tells you the table’s push on a book is a contact force?
Example or sketch Draw the force a compressed spring exerts on a cart touching its right end.	**Non-example and why** Earth pulls on a falling ball after it leaves your hand. Is that a contact force?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Non-contact force

Meaning in my words How would you explain non-contact force to a friend?	**How can you tell?** A magnet tugs a paper clip before they touch. How can you tell contact is unnecessary?
Example or sketch Which downward arrow belongs on a ball after you let go?	**Non-example and why** A hand holds up a book. Which force disappears when the hand moves away?

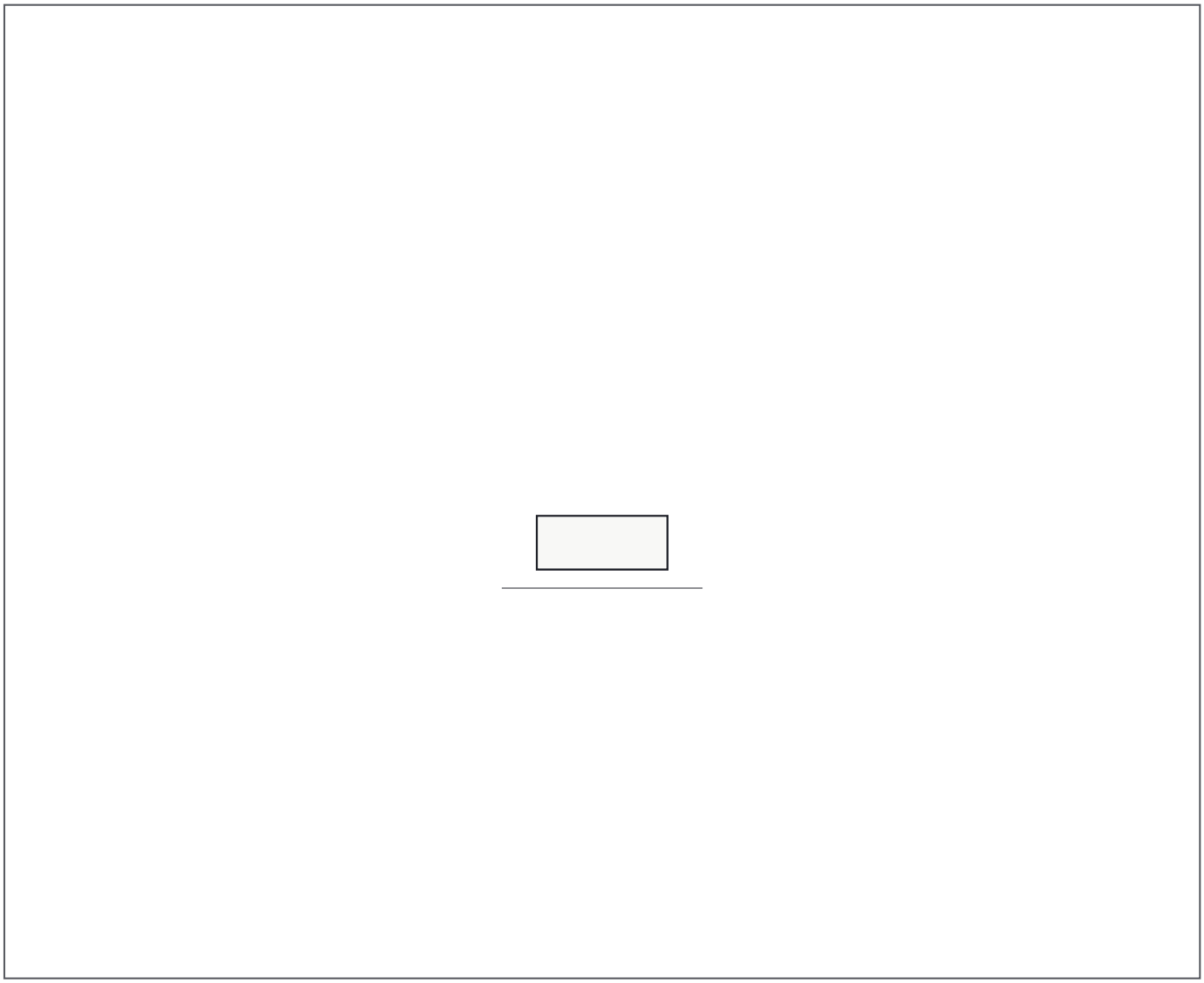
Mass vs. weight

Meaning in my words How would you explain the difference between mass and weight?	**How can you tell?** What changes when you take the same cart to the Moon: its mass or its weight?
Example or sketch On Earth, use 10 N/kg. What are the mass and weight of a 2 kg cart?	**Non-example and why** A label says “mass = 20 N.” What does the unit tell you?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw Earth and a crate. Put weight on the crate; label source, receiver, direction, and N.



My marks mean

Who is doing the pushing?

10

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Name the source and receiver of each force.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

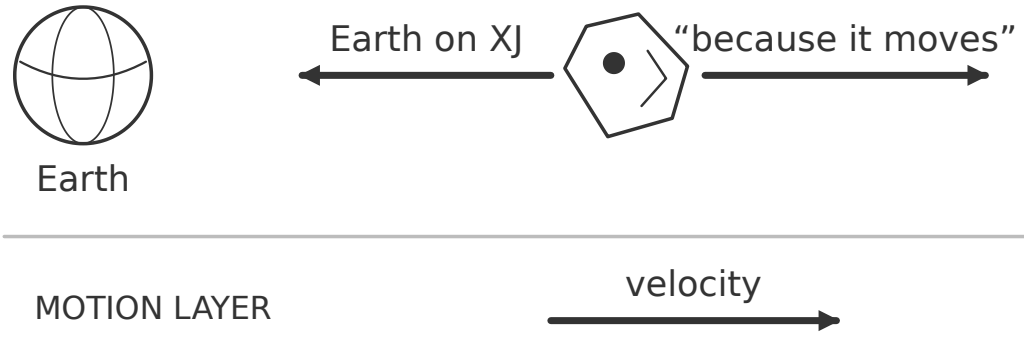
Approve the force layer for XJ. Name who exerts each force and what receives it. Keep motion separate.

SCIENCE GRAPHICS EDITOR ☐

Prepare the viewer graphic. Ask your partner who causes each proposed force before you approve its label.

Point to a clue. Keep your first marks; add the update using the color key.

An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.



01 OUR FIRST CALL

Which force arrows should go live? Explain each arrow you keep or remove.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new information change the force layer? Explain your update.

Switch speakers. Who causes each force you kept?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks